

2024 ReviewStream

ReviewStream | 1:27:42

Okay. Hello and welcome everyone. This is the 2024 review stream. It's December 20th, 2024.

Welcome to the Active Inference Institute. We're a participatory online institute that is communicating, learning, and practicing applied active inference. This is a recorded and an archived live stream. So please provide us feedback so we can improve our work. All backgrounds and perspectives are welcome and we'll follow video etiquette for live stream. Head over to activeinference.institute to start to find out more. And for those in the live chat, we will greatly look forward to your comments and questions and we'll return to them at the end of the stream.

Okay. So today in the review stream, as we have done for the two previous years, we're going to recap, reflect on the streams and video production of 2024 and pre-flect to 2025. In other words, to take a break from streaming and documenting and typing while out there driving through the active fields to take a look back so that we can

be how and why we want to be in the future. So maybe by way of intro or warm up, Dean, where are you at? How do you pre-flect on the reflect?

Well, I honestly, Daniel, think that this is probably one of the most valuable exercises one can carry out. Again, you don't have to have been there every day that another piece of content was captured. But I do think that being able to take one of these, step back and have a look at the entire entire cycle is, it's just super insightful in terms of my own learning. And I think it's, again, something that there is no proxy for you. You either show up on these days, and it you come come away, way smarter, or you, you think you know it all. Until you, until you look back and you go, Whoa, holy mackerel.

Holy mackerel. So I love doing this.

Awesome. Okay, let's just jump in. We'll see where, where we get to. We'll see where we triangulate. We'll see what else people put in the chat. Okay.

So here's a screenshot on the right side of the back end organizing of audio visual production, which is truly something like you.

See an interesting paper or other shared note in the ecosystem. Find the contact information for that or those people. Send them a templated invite, or sometimes a slightly personalized invite.

coordinate the date set up the format that they're interested in. Make a row in this table, and then lights, camera, action, and stream. And that's happened now three, four, 500 times.

So as the sample size grows on, on this and other fronts.

Again, so much of the looking back becomes even the more valuable.

And this year, there were 137 such scenes for which there was also a behind and a before, during, after.

137. And we're going to use that. Obviously, we could just play the 137 videos concatenated with each other and have a 220 hour, whatever it would be sequence.

sequence. Oh, you know, we could just line up every bird in the ecosystem.

Or we can do something that's fun and and we use in the past and the present and the future.

So a few highlights that I expect we will enjoy.

Thinking about how this might go.

About those 137 videos.

One of them was the majority of the active inference insights episodes that Darius Parvizzi Wayne made.

These were basically the only video productions in Institute time.

that I didn't play a role in.

So that's not the only reason, though.

They're also the most popular video productions that the Institute has ever made, possibly because of Darius's charming nature and easygoing conversational style and format.

But they truly were great sessions.

As characteristic as well, we had many fascinating guest streams.

Ranging from the the.

The.

Hurried to the beyond deliberate.

We had probably the two longest streams of my life.

Like that's a don't hold me to it, but but probably above 80% on that.

The two mega streams of the symposium.

And.

Only two.

Paper focused live stream series.

With a full dot zero dot one dot two with the authors.

And.

That was interesting and different.

Maybe we can explore.

Uh.

How that came to be.

With only two papers being discussed, but over the previous.

Three years.

We did 56.

Then.

Dot dot dot question mark question mark.

And.

Math art.

So I'll let you.

Share what did you see in this.

Orthogonality and space between.

Well, one of the things that we, when we do these.

Presentations, even today's review stream.

It's.

It sets itself up.

As you said, we can line up all the birds.

We've got the bullets.

And everything is sort of.

In nice, neat rows and columns and stuff like that.

But one of the things.

That.

If you are.

Able to internalize.

The.

Process of active inferring and.

Consider yourself.

Maybe a subject matter expertise and.

X and.

And.

And be an active inter infer.

Then the question.

Becomes.

So.

Well, what does that mean?

One of the metaphors I might.

Suggest is that.

You.

Move from.

Just being able to.

Pivot or slide or.

Take in.

Each bullet in a.

In a sequence.

And start developing the sort of 360 degree view.

That you.

Already.

Have been afforded, even though you don't have eyes in the back of your head.

You.

You're kind of aware.

Of everything.

In your niche around you.
That's.
There's an advantage in.
Not just being stuck with.
With the blinders.
On.
The math art.
Of that.
Kind of.
Sets up that.
360 degree potential.
Or policy.
Because it.
It's.
The way.
I would describe it is.
If you have.
Math at zero.
Or.
Art at 360.
60.
And then you have this.
This ability to come.
Full circle.
Complete a cycle.
It just creates the kind of.
Gapping situation where.
Going off on tangents.
Isn't just.
You know.
Going down rabbit holes.
So to speak.
There.
There's some real potential.
To.
Gain.
Some sense.
Of.

How the pattern recognition works.

Why certain patterns.

You can take your eyes off of.

Because you think that they're going to be stable.

Why others.

Seem to dissolve.

And percolate up in completely other different contexts.

And you don't recognize them.

But.

Us taking.

This.

This potential of holding up.

To things that are perceived.

As being far apart.

And not having to necessarily build a bridge.

Across it.

But just be able to say.

Okay.

So what happens if you.

Keep an eye on both at once.

It can.

It can.

It can change the way you think.

Yes.

Both.

At once.

It.

It.

It.

It.

Just makes me sort of think about.

The.

Representations.

Of.

Insect vision.

Being all.

Compounded.

And sort of.

Pixelated.

When.

We also have.

Just the.

Minimum of two.

Eyes.

With a unity of perception.

And then.

That even goes.

Outside of our.

Visual.

Field.

With.

Just.

Our spatial.

Situational awareness.

Okay.

Let us.

Begin.

So.

The.

Stream series.

In no.

Particular order.

Just.

Basically.

Haphazard.

There were.

Several dozen.

Guest streams.

Ranging from.

January.

Up through.

Just.

A few.

Days ago.

There are.
Interfaces.
And.
Tables.
That you can.
Sort.
You can.
Sort.
You can.
Sort.
By date.
You can.

The tables.
You can.
Put them into.
LLMs.

Do all sorts.
Of different things.
With these.
aspects.
Aspects.
From here.

From this.
Moment.
I just.
Pick one.
And that was.
One.
Diego's.
What could come before time.
Because I really.
Thought he did.
Well again.
He's the.
A lot of.

Herse rules.

Thinking.

Always.

I've always found that quite.

Interesting.

Hmm.

And that.

Brings me to.

To another.

What is the limit.

Question.

And which of these.

Guest streams.

Are.

Topical.

Like the topic.

Of embodied intelligence.

Which one.

Ask.

A question.

Which one.

Present.

Attention.

Or.

What.

What they do.

Just in the title.

I think.

Some other highlights.

I see.

Are.

Whether it's.

Anna Pereira.

Or.

David Blumen.

Robert Warden.

John Boyk.

Seeing the research fellows.

Before.

And during.

Their.

Fellowship.

Also.

Play guests.

At.

The home field.

And.

I think.

The.

One.

That just.

I'm drawn to.

Marshall McLuhan.

And.

Project 69.

Gas stream.

69.

With.

Andrew McLuhan.

Because.

It.

It.

It.

Was just.

An.

Awesome.

Example.

I.

Think.

And hope.

Of.

Of.

Actively inferring.

Seeing what happens.

Rolling the die.

There was a.

Last minute.

Cancellation.

That.

Otherwise.

That would have been.

To.

Happened to connect up.

All of these.

Dots.

And.

And that was the.

Throughline that.

That.

I still.

I guess.

That's the question.

Enjoy.

And reflect on it.

And learn again.

From.

With or without.

Relistening.

Listening.

Yeah.

You.

You.

You.

Don't necessarily.

Think.

These things.

Should just.

Work out the way.

They do.

But when they do.

It's.

It's like.

Oh.

Okay.

Roberts.

He.

He laid.

He.

He's another good litigator.

Of his.

His ideas.

And he laid out a really good story.

With some people.

Who.

Could really.

You know.

Shredded his ideas.

But they had to.

Sort of.

Keep an open mind.

And even though.

They asked some good questions.

He was.

He was really.

Comfortable in.

Sort of explaining.

What he thought he'd unearthed.

So I thought that was really.

I'm glad you mentioned Roberts.

I really liked his too.

Yeah.

And also cool to see.

In turns.

Just.

I hope we can continue.

With the contributions of others.

In the coming years.

More and more.

To curate and invite.

Broadly.

And globally.

And timely.

And relevantly.

Because.

We can keep one foot.

In active inference.

Through multiple different ways.

And still be in triple threat.

And pivot.

And assist.

And connect.

And reach out to people.

And sometimes it's as easy.

As just present.

And then let's see what the live chat says.

Other times.

We or they.

Want to do a little bit more.

Prep.

To make sure.

There's going to be some.

Some.

Clips.

On that.

Rock climbing.

Face.

That we're going to go on.

Together.

But the guest stream.

Is just.

A really.

Unique.

Opportunity.

For people who like to think.

Active inference.

And.

And then they.

Run into that.

Madlib.

They run into that bar.

So.

Fun times.

With the guest stream.

Okay.

Live stream.

So.

This is sort of the.

The.

The.

The one that's named.

After itself.

Because they're all.

Live streams.

Basically.

But the live stream.

Siri itself.

As mentioned earlier.

Focuses on.

A specific.

Paper.

So.

When.

I look at.

These two.

Papers.

That we discussed.

Which were.

In the first half.

Of the year.

Working.

With Chris Bennett.

On.

Preparing.

And.

Executing.

Active data selection.

Information seeking.

Paper.

Which.

Thomas Parr.

And Carl Friston.

Joined for the discussion of.

And then.

In the second half.

Of the year.

Leading up to the symposium.

Working with.

Arun.

Nirenjan.

On.

From pixels to planning.

Scale free.

Active inference.

Which.

Lance.

DaCosta.

And.

Carl Friston.

Joined for.

These.

Were.

Heavy.

Papers.

Every paper.

Is different.

Some.

More so.

And these were really.

Heavy.

Analytically.

Complex.

Papers.

In certain ways.

And yet.

In that.

Ever so.

Delightful.

Active.

Inference.

Way.

People.

Had.

Intuitive.

Grasps.

And.

Resonances.

With.

Active.

Data.

Selection.

Whether.

Eye.

Sacade.

Or the.

Machine.

Learning.

Case.

Is about.

Looking.

Not.

Just.

Information.

Gain.

Is going.

To be.

The.

Highest.

And.

Understanding.

How.

Is that.

Structurally.

Set up.

So.

Down.

the case especially if you're thinking about scenes that change in time or other situations. So to work with Chris and really go into okay how does information gain epistemic value have similarities and differences from maximum entropy? Those are some core connections to make in active inference and so to be able to climb that hill and and get there and then come away and now look back on it is all part of the learning. And then similarly in the from pixels to planning paper which was possibly one of the splashier in our field overall this year it really operationalized a lot of the developments that have been happening in multi-scale modeling and connected to the renormalization group methods which two three years ago were not so much on the table for active inference at all and now they're being taken into and across different domains as that paper works through and so for it to dovetail in the dot two during the symposium and just it was great. So two thick papers with a lot of direct and accessory learning both done with very mature and diligent colleagues and to have Carl and a plus one join both times. So great times fun papers and anything to add on that Dean?

Just a thank you to all the people that participated in those because it's one thing to write a dense paper it's another to show up and do your best and to try and support our community and sort of take it off the bench and try to be able to explain it in ways that people who maybe don't have that same degree of background might enter that was the one thing i took away was they were they were they weren't dumbing down what they were talking about but they were also going to great lengths to make sure that people weren't intimidated by what they had to say that they felt that it was something that they wanted to not to share but make accessible.

Thank you yeah it's a point we we truly can't make enough which is i and we are very appreciative to the all volunteer cast of this show and this channel and this institute and ecosystem largely so for people to write the paper it is easy to put it in your rear view mirror meaning put it behind you put some distance between and never look back but to add value and promote open science and to engage with the true serendipity and spontaneity of the live chat and the live stream format and again there's there i can continue on this angle but through undergrad and graduate school i saw so many paper-oriented lab meetings so live stream format and then guest presentations like guest stream format good points were raised amazing points perhaps novel points perhaps critical points for reproducibility side insights deep lore like incredible people dissipate into the wallpaper never to be seen again so not that everything needs to be open all the time but the start streaming and stop streaming button gives us that in and out and so i i'm just really thankful that that so many people people in the research area and increasingly in applications do step into our clean well-lit place to discuss active inference and that so many so many participants in the live really live and in the follow-up sort of online discourse area uh get to engage with but also like co-create and even lead some of the research so that's just too epic not to mention all right there were four round tables quarterly round tables the last of which being yesterday but the date's just off by one here so check those out i i could have screen shot it again i could have put every picture of every bird we could have put every frame of every video so just showing more by showing less four round tables were great i look forward to this quarterly tradition it's part of how we see nested time scales from the most tactical and operational and day to day through the monthly newsletter and the quarterly rhythm of the round table the annual rhythms with different cohorts turning over symposium all these different things and many more to come so it's just awesome to to compile these updates and it is um a learning experience not unlike a reflection to feel busy and feel that progress was made and there's places where progress wasn't made um windows that closed and uh or windows that remain open and yet the inertia and the challenge of actually

preparing a meaningful reflection because it can feel just like studying history

but those who understand the past control the future

any comments

all right you sounded well okay or extreme so this was the three videos that were looks like in the beginning of the year in january march and april four nine two squared three squared and these streams in different ways dealt with organizations two of the three with sense janita and hazel and with nations nathan schneider were from the metagov community so that was really cool to just connect with our metagov colleagues and

and again provide a space that's like a two-way street there's people who are going to be familiar with governable spaces or not and active inference or not so all four of those quadrants are especially in play when we go outside of the sort of friston co-authorship space and and that's where there's these these sparks that can fly and in or extreme seven with jamie joyce and john ash great discussions again about how in small and medium and large groups in our modern synthetic augmented environment does prediction attention memory all these cognitive phenomena that from the research side from the high road side

we're talking about unifying and what do those look like on the ground and today

okay

all right model stream so here in a sort of countervailing pattern to the live streams where there's the 56 over the previous three years then followed by two here we can see by the numbers that there were nine model streams over three years and then eights over this year so model streams and some of this can just be categorization because each stream only goes into one category so it's it's it's just again just remarking on the

the way that the cards did fall not not the way they could have fallen um the model streams go over specific models whether they're active inference specific models or even a little bit more general like a pomdp a partially observable markov decision process it doesn't have to be explicitly active inference based but that's how we're coming towards it um these streams from the math and the computer science

side add a lot of primary material

and that secondary material layer like follow-up questions or just the order of presentation from these experts can can get the the much larger crew who's at the third layer to understand what these models are about so for there to be this sort of air of rigor with informality and engaging the technical details and just seeing how the presentation takes off and and it's like all on the modeling steady hands and then the presentation's over but we're still there with the informality and and like that after party for the model almost which is like really where the connections are made and and uh the one that i would highlight was ran ways work on the value of information and reward specification which kind of connects back to the active data sampling too and just this idea that well we have mdps markov decision processes let's just say like chess and then we have the po mdp like chess with some observation error that's the partial observability part and then the idea that we can see the pomdp as an mdp where you're observing an uncertainty measure and so that connects a lot of dots with

reinforcement learning and what epistemic value is it's not epistemic value towards future pragmatic value
it's a separate source of value and there are situations where those vectors
cancel each other there's situations where they're orthogonal to each other there are situations where
they're doubly aligned and to be able to really neatly and concisely see
here not that this is the only way to frame this but here's how active inference differs from
reinforcement learning in reinforcement learning you have utility function you need to coerce or shoehorn
epistemic behavior heuristics whatever you want to think about into utility so you can use all different
kinds of mechanism but you only have one currency that's utility or reward and you're reinforcing and
your training on that epistemic value is the second eye opening that's what makes it different having
the epistemic and the pragmatic value in the expected free energy equation not just coercing the epistemic
into the pragmatic so what a great point what an analytical connection to lock in and also a jumping off point
for
thinking more broadly about modeling how much jumping off do you think that the authors of these papers
in the after party actually
materialize
materialize
because it's that emancipation piece that i think is
is really important if you're using it as leverage how much
that do you think
materialize
hard to say sometimes finishing a paper it feels just like
pulling up to the side of a pool and being exhausted after bringing it over the finish line but then i think
i think
then there's a rest and then
they
push off
in how maybe they see their next research dimensions going
and and i mean let's let's see what others think and so on but
then it's it's more like
the model streams are those are those are the
those are the hard ish crystal parts
floating out there so
there's this sort of like history of science like
how did how did jock
jump off after this paper i think that's really cool and interesting but for the other eight billion
people it's about how they will push off of this paper
okay
all right math stream

so there were

five math streams

these were all

excellent

richard

chervijan is a participant at the institute and prepared for his lab group

as a

part of some of the completion phases for his phd

this two-parter on bayesian mechanics and sort of

re

gave that lab meeting presentation on these streams i think that's a pattern we could see more and more

people going out doing stuff and then coming back and giving round two as a dot one or or some other combination

with jonathan gorard in march this was uh

one of the most watched live streams of the year which was great to see because we laid out a lot of connections between

stephen wolfram's observer theory and a lot of what was happening in active inference

so there was sort of this like funny thing where i i just saw again only one way to

see it like this but just there were conversations like on machine learning street talk between

carl and stephen between friston and wolfram

then it's like

that is a splash that ripples

then

there's kind of the postdoc level

discussion

so that's just one like academic structuring to see it but it just this is how through time

it's like a zipping

and and an annealing of different research fields and how they co-inform each other

and the way in which we're going to be able to use research tools to translate between ontologies

notation systems and all of these things is going to turn an amazing spotlight back

and that's going to reflect a lot forward too

thomas varley gave a great stream on information and then

dean one of our favorite colleagues toby st clair smith with the sort of um live stream 54

this is not 54 follow-up on his new structured active inference which both addresses many critical topics that the computer science side are thinking a lot about like how do we do active inference

with apis how do we think about multimodal interfaces and active inference all of those questions

and the category theory and his charming ineffable style

any comments on these no

okay

there were textbook group activities from february on through i believe october
um so they were sort of like semi live streams because they're live while they happen but they
don't directly go out to the youtube channel but then after the cohort has finished we flip them all to
open but they're unlisted during the cohort um not sure why but just don't don't want to string it
along want to release the full cohort recording at the end um it's now been
in several years since the march 2022 release of this textbook we await more and other textbooks in the
time to come but there's something nice about revisiting a text not just looking for more and different but
after reading each of these sections so many times and still finding new familiarities and new unfamiliarities
it it really
highlights and and thinking about what what is the role of of an attract text as a tractor
text as entry point text as jumping off and
how do you think we can well i could i could add more about the updates on the group itself like
about sort of the participation trends but i'm just thinking here you know what is the role of textbook
groups that sort of learning pattern in how we how we welcome people into our yellow learning circle
i think what's really interesting is is that um you you'll you'll have more to say on this than i do
because i don't directly participate in the textbook group things but if people have a chance to have
conversations with one another on the same subject i.e this book and its contents the this is there a is
there a greater potential to open up each one of those perspectives to that 360 degree camera view i mean
you've you've gone back and looked at the book multiple times how many times did it take you going back
to your 360 perspective to to become really tangible
i don't know if it's got more or less tangible through time okay um
this year andrew pesheya co-facilitated a handful like half of the meetings so it almost became more tangible
when i see it from the side
and see another and i just think wow he just laid it down directly that was like laying down railroad
tracks whereas i just feel like i'm sort of a lightweight forager leaving a little pheromone trace but
when when i see another um laid out it's really different even if it was semantically similar
um and then to the 360 that when when you mentioned that um you know 360 um makes me think about like
a wedding or an event and then it's like well where where'd you come in from or what vector did you come
in from and some people volunteer that information other times it comes out indirectly but then
if we can have that theory of group and and realize oh people are coming there was a 360
that came to here so this this is a ready willing and able being approached from multiple sides
embodied only and specifically by the actually presence and for those who weren't there it just it wasn't
that way for them which is how you can say things are in general you know it's it's because it's
interesting to me if if um if the pattern holds most people that are that are are are taking up a text for
the first time they will they will tend to follow the sequence of the chapters the 360 doesn't happen
until they are able to say okay what happened at five o'clock means something different than chapter
five what happened at two o'clock it's now it turns into a question of relativity as opposed to okay so

i i completed chapter five how did that closure now organize and set up for the for the next mini loop right on the on the continuum so it's like there's a bunch of beads strung together for the first time engagers whereas the 360 viewer could quite literally step in to any chapter and because of their priors now their their their free engagements with the book they you could you could literally go i could say to daniel okay uh i'm thinking about this thing from chapter six and he and he's already seeing it nested within the entirety of the book it's just a question of when does that emerge so that's the part i find really interesting because i'm not anti-text but i i also know that most people end up stopping as you say and things get yellow after they conclude the epilogue yeah i guess just briefly to to kind of connect that to some of the specifics of the textbook group as through many discussions we've come to see it we know joking about reading it forwards reading it backwards middle out all that it's like well chapter one and ten are the bookends they're like the opening and the summary then you have the first half chapter one through five being in as they frame it more epistemic and the second half is more uh application oriented and it's like zoom into the first half well you got chapter five with the neuroscience and reviewing about where active inference has been applied to the system that so far has been studied the most the mammalian neuroanatomy and then you have the act inf in general in two three four and then you have the generative model four and the high and the low road in two and three and then going back to part two there's the recipe to begin it then there's seven and eight continuous and discrete time and then nine with the data it's like but if i just had this stare at the 10 chapters i wouldn't have been able to memorize that but then going into it so many times just cycling through many coats of paint many spins the tech being okay with not knowing and then it's kind of like seeing each chapter as an interface and all of a sudden there they are all at once they tend to morph into rules as you as you gain familiarity and that's that's why i was i was curious when i asked that i'm always wondering whether the person who i'm talking to sort of catches themselves moving from that that content context as as the as the you know the the shorelines and moving into those conceptual rules-based heuristic stuff it's now internalized to me i can go i can go in different places and not lose the thread yeah golden thread golden ring okay

fourth applied active inference symposium so wow what a what a what a fun ride this one was um it was in november there was a third day session that john clippinger led while i was blissfully driving away elsewhere um but on the 13th and the 14th there were these two mega sessions ones let's see here's the first session um yeah here's the 19 segments on the first session the the the pre-recorded ones were concatenated at the end so that was like cool but but the uh the front of house back of house back of house on the live handoffs was i mean it's it's fun like if someone doesn't think it's fun they're probably right but if they wonder if it's fun i would recommend that they try these kinds of things um and then here was the second session but these were like 12 or 14 hours each so again it was it was bizarre i i exercised a ton and just listened and learned a ton so and and uh but certainly was in an altered state of just learning and being focused but also very distracted you have a new definition of how to manage the binge yeah i guess just put it into do a weekend and then and then i don't know so it was awesome i mean again we say say more but people want to know how active inference

can be applied and we coordinate this symposium and and seek curious co-organizers to get out in front for next year let's start in january or do something amazing for number five um maybe could you just give a remark on how i don't i don't like use them use the mean of tipping points but i think it's i think it's really appropriate i honestly think that just just getting to that place of no return and and seeing that things will continue to self-organize like there's a huge element of that you you a role that you carried in terms of trying to orchestrate things and that that's people don't necessarily have to know the details around that but there's also a huge element that this thing kind of has an element of what we talk about all the time which is a sort of self-organizing system and and watching something auto poetically get its act together um i think is again one of those one of those opportunities to learn how processes there's there's there's there's a certain amount of hands-on to it but there's also a huge element of hands-off and then it's again playing in that space that um gives you confidence that that would be my only comment cool yeah that just makes me think about carl and the the panel with alex auroria and that was like the only one where there was some technical um comedy with the with the jumping around and all this other stuff um and and yet also it was very it it was like they had co-written mortal computing and that was also a major contribution of the this year and and uh so it was like it was kind of like abbott and costello back again but but it but it was the after party or something like that so it's it was awesome okay fun times morphstream so this was a series that sarah hamburg proposed and did the other two last year and she did one stream this was uh it's a great topic the connection of morphological and and neuromimetic computation cognition and active inference and she did an awesome job with this morphstream and

i look forward to people continuing to add more rows here okay now we get to active inference insights what what do you want to add here well i i did a screen grab of of um the it it's not ironic i think i don't know what there's been any any empirical research done on this but as people are trying to gain insights they're looking off into the distance and i want to catch darius and his and his great locks something i'm envious of staring staring off on the horizon because i mean bottom line is uh he stepped up and he he i thought he did a pretty polished job of of bringing a whole bunch of people a cast of characters into some sort of coherency so um my my doing screen grab here is out of respect i'm not trying to make make fun of this because i i do think that there was a a certain amount of learning on the fly that that darius wasn't afraid of showing the audience and i think that message came through loud and clear and i i really appreciated that there's a certain amount of vulnerability that i think everybody that wants to get into this for the first time has to you know sort of carry into the situation and i thought he did it without sort of pandering or or you know acting like this was overwhelming whatever so good good balance i totally agree i think there was a really um prodigious balance between the preparation that clearly he put in both in general to upskill in the field get familiar with the late breaking changes in literature and all that and then also clearly before each specific interview and yet to do that background so that he could look off to the side it's like the anti-teleprompter the background is when somebody's actually put in the the preflection put in the background so that they can

push off of that firmness

and and also for that moment of openness and uncertainty

in linear online time to be something that people empirically like to listen to

so that's that's an amazing balance and yeah the guests were great and we had a little reflection of

our own where we switched positions for that last one but the rest were just him and the guests

and they were the most watched videos on our channel and it's uh challenging for me to

not have the visual because there's such a visual element so to do it as a conversation which was his

intention from the beginning was to make a conversation style podcast style not and here on the left side

you'll see this that's that's challenging because it's it's big topics so just to have it strung along

it can from an expert's ear make a lot of sense but but how does that become enticing and accessible

to somebody who may not know why they're putting a b c x y z and if you want to drink a game every time

he looks away just take a sip and it's it's actually quite relaxing every time he he humbly brings up

one of his papers or thoughts don't drive after that

um okay art stream so we we're we're we're starting to circle we're starting to to approach math art but

first we're we're stopping in in the pure math um stream earlier now we're here in the pure art side

again these these these um categorical distinctions the categorical imperative they're they're almost

their own form of of sorting and and like this year starting the art stream series

you know saying we're not going to put order and change in art into a guest stream we know we can

but what what if we open up an art stream avenue what if we had a separate link so that we could share with
our artistic colleagues to let them know that we have a lane and a channel and a touch point for them

so that is fun to see a new series start again we could just call we could just number all videos

sequentially or we don't even need to put numbers on the videos at all so what does the number really do

um all of these were really interesting the one just a few weeks ago with corinne kukkonen was

fantastic the way that it just dove it it was novelistic and novel just to begin it's like you

open the book and you're right there you know it's on the street and this is happening right there it's

like you're in media res and the way that she

brought us in media res into this higher order predictive processing reading that doesn't deflate

or fall into any other number of sort of simplistic fallacies or or ways to constrain the variance and

the open-endedness of literature but just engage so richly with the material substrate of the text

and and and um bring in cultures and angles that we know we've never seen on a stream before

and that's one of the things i really like her presentation too one of the things that i

kept thinking about was she was talking was i think most people think they they get into a book

right like it's it it draws you in and and she she talked a lot about how it can also

it can also free you free you up so help you get outside of your your patterns which i thought was

actually really quite fascinating so that part of it could in my background i was an english literature

teacher and i never had i had access to the information she provided i think i would have gone

at presenting literature in some maybe a little more novel ways so that was a nice surprise yeah cool okay

math art

starting in only may with math art stream one and completing the decade in the 10th month

i'll let you take the first pass on this

well i guess the first thing we should do is acknowledge our leader uh the the i don't know if

she would be a queen or she must be higher than some royal rank shanna um is so ready enthusiastic and

she really does have this ability to take that wishbone moment and not get stuck at the branching

point like she can take the whole thing in at once and really appreciates anybody who is willing to take

and back to the risk thing and back to the risk thing take that risk and look at things in

a variety of different ways so hector uh

magdalena uh sonia sonia came as a as a guest and again all of these people that are trying to

to not get cubby hold into there's only one way of looking at this um it takes a lot of enthusiasm and

it takes a lot of um it takes a lot of community and i think that that was one of the most enjoyable

things was when jim and shanna and estelle and when everybody was sort of they just they they they made

it feel comfortable they made it feel i'm not a safe space guy but it they wanted people to take risks

and that was you i don't i don't put a price on that i just think it's invaluable so

yeah thank you for that and a great appreciate the the facilitation and the the holding of the

creative center of shanna and hector and everyone i greatly enjoyed math art stream 4 with kirby erner

on dimension and synergetics because it was like such a world weave moment for myself and a learning

experience to see dimension from this other way inside and and it just it was that multiple ways

at once visualizing the fourth dimension and it was all about one two three four five and that's the

traditional or at least mainstream way that people think about adding dimensions and that's the computer

scientists data structure and that's the the brain bending question of the fourth dimension visualized

for many and then here's cool kirby just like hold the tent in your hand hold the grain of sand in your hand

there's the four it's like what do you mean there's the four it's like they're just all there at once

spatially and here's the math so i think it's awesome and uh then the can i say one thing apologies to kirby

i actually put this image on this slide in dedication to him and then completely forgot to mention his

name as you can see tickles 24d right so yeah uh shout out to kirby nice cool cool kirby my my the closest

thing i know to a human Yoda and then here is uh math art stream 11 the mathematical holiday poetry party

we didn't realize it when it was planned to be but i was thinking like you know there's people who have

canceled streams weeks before or minutes before there's times where we've rescheduled but only

shanna has taken the non-stream and turned it into into its own performance of like this dark like the

street it hasn't happened but it's like now especially it has so that's why i have it sort of like fading

because when we imagine in in the dark imaginarium the in the dark symposium if we imagine it it's like

it has been because it hasn't but it hasn't because it has only been imagined and so when when

we're playing with so many things about what what what is cognitive and real and embodied and all these

it's it's just a symbolic tour de force and again for most people they just like cancel

and that's kind of the the it and all but but even a cancellation goes further

okay okay active inference stream i'll let you take the first pass on this

no i i can't make any passes my my putting these two images on here is because the uh the shift the eyes

on the end that's the extent of it most of the stuff you talk about i have no clue what the heck you're doing yeah well this also began i guess just four months ago but i've really treasured it as an opportunity to give like a little topical slice into some development effort also heavily using cursor and llms and code and all of that and i i i try to leave it all out on the dance floor i try to bring in the the deep baseball history and references and skip around um i i i hope it's enjoyable and kind of like a variety show with with all the shiner when you've left than when you were wrong it's it's it's super funny and and a a a a topic that we always come back to with okay there's subject matter expertise and what and just seeing like literally over these streams you'll see even in the six seven eight and i'm saying it's doing 800 lines but 200 lines but 200 lines was the limit just a few months ago so being there when the tide pulls out and the 100 foot wave comes onto shore and being like no the subject matter expertise and the cramming textbooks of bayesian statistics into your head and the coding and the for loops and all of these it's like like that's not going to fall onto a human's shoulders in most cases anymore so as that as that knowledge playground just morphs i mean these these may be seen as some of the most rapid months of synthetic intelligence acceleration in a long time yes in a 10 000 or 100 000 years can you say that in history like it's it's we're on the you're on the almost vertical like i i i think there's that there's the sort of fractally element of zooming in and it being a hockey stick oh canada um all the way but at the same time just just talk about seeing it play out on the day to day with the new versions of cursor and the new language models and and the the the fixating salience of the sophistication of the technology technology and like the kind of promethean angle on that and the the the power and the potential and open source and revisiting a bunch of previous papers that that i've written qualitative ones collaborations more structural ones kicking it off with the field shift just saying hey you know it's not necessarily you might not give an a plus but at the at the b plus to a level we have the all by all maps like the sphere of the shifted like you could think about a in terms of b it's like okay well now we have a through z by a through z so now that the the board has been sketched right and that that that's just like it's daunting but it's empowering but it's terrifying and it's also exciting and so what what does it mean you know when when careers were made on what if we thought about a like b and now we have that low bar rising tide will will it float all boats and and what is gonna get swept away when that technique becomes increasingly powerful to someone who has zero experience with how you play with each of these different tools the thing that the metaphor that keeps running in my mind is it's not a roller coaster that's already been set up in the field it's it's quite literally the rails of the roller coaster are only about developed only about 10 feet in front of you and it looks like you're going to go off the rails and just you know fall to your death but it just seems like there's more rail as you progress there's more rail literally materializing as you move down the tracks and so yes that is terrifying yeah that's all that's all i'm gonna say yeah and just the who's on first dialogue the who's on first dialogue light motif just being funny and information laden for me and maybe for others but but again just pointing more broadly like whatever cultural vehicle or format you want information in that that roller coaster is going to take that turn in the next 10 seconds you know the book is there it's

static and there's there's so much to be said for all of that too in the shared reference point of the text like we were exploring earlier but when when the book is morphing second to second you can flip back to a page it was like wait i just was like it's like it's already different enough when the text is static and and it's like being on a on a hyper mutating trampoline so so just seeing that happen and just it it just how quickly we've gone from here's a great blog post you might want to check it out they use this cool metaphor to like i'm 75 prompts deep like wired into this alternative reality game but i'm also getting this like tacit understanding of bayesian statistics is like you know will will again just speaking personally well i have come to see will i come to see the hours with the organic chemistry book in the library and those years as the ways that are creating the foundation of habits and ways that become empowered will those study methods and ways be misleading or worse is it something that others just analogously would want to engage in for like the character building or for the traditionalism or is it just like a historical legacy cope and and new nestmates will do things that will just completely you know it's it's probably some of all of that it's just that it's it's evolved so much and i don't feel like i've been studying that long but i can't even there's no venn diagram some days between when i finished undergrad in 2014 and how i was studying and where it got to this year so here's a question for you like a couple of years ago 2022 2021 we spent a lot of time in the institute pondering this idea that it just seemed like markov blankets were arbitrary but because i'd done some programming in the previous decade i had equated markov blanket and where where they're placed with the idea of so what's good enough it's not just arbitrary it's i don't want to get caught up in a recursive loop comedic outcomes aside i know who's on first and that's good enough for me like i don't have to sit there and continually ask prompt after prompt after prompt right so where do you think this idea that a lot of stuff that we do know now about active inference where is it going to um bump up against this idea that a lot of information now is going to be synthesized through these llms but not knowing what the llm's definition of good enough is relative to our own yeah that's a question to revisit it just makes me think of of the hand on the page and there's this this chain of continuity materially from like the ink of the printing press on through your fingertips and and neurons and glia so there's like this material chain and then now it's like we have a whole team of telephone between the whole library and us at the terminal so there's so much obscurity and yes with open source models we can look a little bit under the hood and all of that but like just do a single dimensional linear regression and check how interpretable that one is and then use that to for a gut check for what kinds of interpretability in in the short and maybe even longer term we can even expect from these massive systems with with billions of components and numerical representations and meaning all of that and and so there's just so much space between that it almost like obsoletes the concept of space between because it becomes its own place to be and i i i guess how can we see that as just the new water that we're in um and and but but but again just like

we should add the um the car meme swerving off the freeway but that's gonna be like yeah that's that's gonna be the the the the educational question like when i see educational organizations grappling with whether using at this level or that level is augmenting learning or a referral to student judicial affairs it's like that that that there's going to be a chasm growing that that is is not going to get taped over simply well 10 years ago i had people that were coming to me and insisting that i i imposed rubrics and i 10 years ago i said those are already they've lost their shelf life and i think everything that you just pointed out confirms that anybody that's still in that way of thinking um um if if you believe that that's it the way that you you measure and qualify what good enough is think think again it's it's that isn't going to help you anymore so cool okay last moments so we have of course one review stream this year happening right now and so i'll bring in a william blake quote and then you can convey what you added eternity is in love with the productions of time what so i just said where that works so so where does that where does that where does the time that's been processed what's going to be done okay so and now behind us then fit in to the idea the concept of scale free because blake i'm sure you you're the blake expert when when blake was unshackled from the asks and the tasks that he had to do to put food on his table how would you suggest he looked back on his day before he broke out the quill and and said something profound yeah well what what a what a big question to to explore the the multiple of his works deal with this in media res realization of fragmentation through memory of a prior state and the the instantaneous primal recognition of temporality of temporality and time boundness which is time as produced the productions of time which whether you apprehend this through reason faith axiom principle framework it comes from eternity of eternity and so this aphorism it it captures that like there's the eternity of the sentences all meanings at once the sentence as one at once and then there's the the linear sequential productions of time and the letters in their ordering and the words in their ordering uh it it uh so how did he see the end of his day that's an interesting question after after what what by by many modern standards would be grueling extremely challenging repetitive etching work it's an interesting question again when i first saw this i thought to myself okay so here we go again we've got a highly respected thinker and he's essentially in his own words saying well there is a an eternity capacity a carrying capacity a scale free component component and then there's a scale friendly component right the cycle of the day is is done what when i'm having my own little moment now to go okay was i productive today i wonder what what what he fits into the well here's the here's the here's the things i made but here's the here's the new

ideas that i'm i'm still i'm still forming and hoping to to to bring into a a way of getting others to maybe go long yeah from from the production of etching and ready to slip and shanna talked about memory and and hector and sleep and how you have to remember to be able to fall asleep um and then also here's just this is from uh uncle john's band by the grateful dead but the last line there where does the time go it's like yeah okay if it flies like from or to where or is it just all at once that's the sort of time crystal time diamond spatialization of time but spatial as a metaphor for moments existing but what would it even mean to exist quote at once when you're talking about something that changes okay last one so yours alone

well i think in 2025 something interesting is going to happen and i rather than taking all of the items on this slide and somehow being able to insert them into a a prompt field in an llm maybe that's possible now like i said i i don't know i don't know how you'd put i know how you string words together in a prompt field but i don't know what would happen if you put a clydesdale a radio flyer wagon a king's coach a miniature horse a table um two two tangential sandwiches and a list what it what it would come up with but i think an active infer if i said to an active infer

rearrange this reformat this

take this and make it a make it a make it your self organize it into your coherency and then and then if you want to give us a narrative but just go through a process of acting in a furry without without you know i don't know how many times you have to re submit the prompts but i i know that it's it that's an exercise this is a different kind of exercise because it's taking information and allowing person who sees it to reformat it and i think that that's something that's going to come to the fore within the act i hope it's going to come to the fore within the active inference community in the the next calendar year because i think this is a this is an ability and affordance that we have to take seeming pieces of of information and not just get stuck in the the sequentialness of it there's there's that there is that but there's also this other way of being able to look at things and i think together that's the potential of not being replaced by ai but actually having it act as a tool in in the context of having this other way of being able to look at information and not necessarily have to continually give it more more definition more terms more refinement it's it's literally an opening up the option is yours like if i said to you daniel play with this and decide whether the where the list stays where it is or whether you take a word like subjective subjectist and put put put it over the miniature horse or which is which is the active inference which is the instrumental like it's actually fun you there's nobody telling you how the puzzle gets completed it's the actual process of doing the self-organization that's that creates the narrative and the generative model that you want to carry into this next year so that's why i put this on here because i think that's the pre-flex moment that you know four years and now the year number five as you mentioned before we came live here i think we have that opposable digit moment and uh rather than being taken out by ai and machine learning i think it's now it just seems like the the moment of sensitivity where we can take this other way and and put the two put the two together

wow okay i'm going to read a timely question from the chat and i think you'll have a take on this one
monty wrote how does active inference explain the bunny rabbit interpretation
uh with the duck is with the rabbit that's how active inference explains it it's as you as as you
uh change swap perspective and foreground and background the what the what active inference
explains to us is that it's the fact that our our ability to cohere take the duck and the rabbit and
see them as with not necessarily one before the other that's one of the great um powers of being able to
not get stuck just in subject matter expertise but be able to predict not just the limitations but also
the leveraging points yeah so if we see time as moment by moment which is the the the commentary
for the baseball game it's a duck no it's no it's a bird it's a plane no it's a duck it's a rabbit then
and then and then that's how next steps occur in contrast we have eternity which is multiple moments at
once and there's space between we have duck with rabbit and the duck with rabbit is in love with the
production of the experience of duck than rabbit yeah yeah and that and again at first that just
that may sound to people like what the heck are they talking about but again if you can relax your
grip on no it's a duck no it's a rabbit no it's like if you can let go of or even if it's just for a little bit
and just kind of go okay well i don't see a duck rabbit or i don't see a rock or a dabbit that's okay
because you're still able to see the all
aspect of that in addition to that oh now it's a duck now it's a wrap nice
horse by duck rabbit by horse there we go but so one of the one of the critical things here i think
is that i think a lot of people will go oh but we live in a world of of hyper trucks now no it's still
a minimum of two if we're talking about the the natural component of this and not just the technology
we're talking about the real and the instrumental so again um i just think that 2025 that opposing digit
metaphor is going to really come into play here okay just copying this in we we can come to closure
but just monty wrote um what is driving this change of perspective the solidification of recognition
reminds me of the process of gaining hand independence in piano um well i i guess that
depends on on the person i mean i think there's going to be some people that are going to be doing
it out of a sense of fear maybe some people that want to be just curiosity they just want to try
perspective swapping because maybe they're feeling like they're in a bit of a rut in the you know the one
size fits all view that they've carried forward for whatever length of time i mean there's almost
infinite reasons why but i think the idea in in a world where it appears that a lot of the
choices are being taken away there's always a reaction to that and people will want to
be able to if there's a way to open up and not not have something make them feel as though
they're no longer relevant so last question monty asks is will the final answer to the fundamental driver
the final answer no it's just relative to the what you can and can't do
do you have any closing comments
i again i can't i can't i can't i can't pound the table enough and hard enough to say if if you
if you sometimes don't want to participate in one of these exercises because you weren't there for
everything um you're you're kind of missing the value in it i i would suggest that this year i was
actually physically present for 18 to 20 percent of the stuff that went on and i got a chance to

because of the nature of the availability in the library of videos go back and look at another 60 and there's at least 22 percent that i have no idea what what went on but i still like to come here because it gives you that sense to step back um so i hope people that are on the discord i hope people that are in the community i would like to see this sort of exercise be something that became more of a i guess there's some way that we have to digitally feed people because if we could get them all into a big uh smorgasbord and feed them while we were doing this we could probably get more people in this in this room but i i just think it's a i think it's a really valuable exercise and i hope other people want to join with you on this next year thank you i'd like to acknowledge upcycle club making a gift to us he said thank you for all the hard work and also jeff so what a stream what a year thank you all see you next time is

Thank you.